Freeze, Don't Perturb: Frozen Generators and Phrase Selection for Plausible Unsupervised Rationales

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Abstract

Unsupervised selective rationalization trains a generator to extract a rationale and a predictor to classify from that rationale alone. Noise injection (Storek et al., 2023) replaces low-importance words of the rationale with random words during training and makes the rationales of fine-tuned BERT generators more plausible. On the USR Movie Review benchmark, we find that it does not steer the generator away from the words it replaces: no gradient carries the noise back to the generator's choice of tokens, the selected words are hit by noise as often as randomly chosen ones, and every fine-tuned generator we trained selects more stopwords than a random selection. Its benefit varies strongly between seeds, with or without a defect of the released code that shows the predictor the word before each selected one. Rationales are plausible only in the runs in which the predictor no longer reads a code in the generator's choice of tokens. The published numbers also correspond to keeping the *end* of reviews longer than 512 tokens, whereas the released code keeps the beginning. Freezing the generator's encoder, which leaves 2,307 trainable generator parameters, prevents such codes without noise: its rationales are more plausible on average than with noise injection, with or without the defect, and far more stable across seeds, at lower training cost. Ranking tokens by their attention averaged over a five-token window, which selects phrases instead of scattered tokens, adds about three F1 points to every generator. Under the published protocol, the two changes raise token F1 from 33.3 to 34.2, IOU-F1 from 38.4 to 50.9, and accuracy from 87.9 to 91.8.

1 Introduction

Selective rationalization makes a classifier inherently interpretable: a *generator* extracts a short rationale from the input and a *predictor* classifies from the extracted rationale alone (Lei et al., 2016). Because the predictor never sees the rest of the input, the rationale is faithful to the prediction. Nothing in the training objective, however, asks the rationale to be *plausible*, that is, to convince a human of the label (Jacovi and Goldberg, 2021). Without rationale supervision, generator and predictor can agree on selections that predict the label without explaining it, such as punctuation or function words that encode the class (Yu et al., 2019), and the problem grows with the capacity of the generator (Jacovi and Goldberg, 2021).

Storek et al. (2023) address this *degeneration* with noise injection (NI): during training, words of the rationale with low TF∗IDF are replaced with random vocabulary words before the predictor reads them. They argue that the noise keeps the generator from learning how the predictor will read its selections, and report that with NI, a fine-tuned BERT generator (Devlin et al., 2019) selects rationales closer to human ones than a frozen generator does on their USR Movie Review benchmark (token F1 33.3 vs. 30.0, IOU-F1 38.4 vs. 29.9).

We set out to improve NI by adapting the noise level during training. We could not, and this paper explains why and what works instead. In BERT-A2R, the architecture NI is applied to, the generator learns only from an *attention-based predictor*, which classifies the review from the generator's attention. From the noisy rationale, the generator learns only one thing: whether the *rationale-based predictor*, which reads it, guessed positive or negative. This guess does not reveal which of the selected words were replaced (§4). Accordingly, the words our generators select are hit by noise as often as randomly chosen words, no noise level or controller we tried changed this, and every fine-tuned generator we trained selected more stopwords and punctuation than a random selection. Noise injection still makes a fine-tuned generator more plausible, by 2.4 to 8.1 F1 points in our

four paired runs, consistent with the noise making a code in the selection unreadable for the predictor; in two of three seeds, the code survives the noise (§8). In the released code, the rationale predictor receives the word before each selected token during training. With this defect, NI *as released* breaks the code in two of three seeds, but its plausibility varies almost as much as that of NI *as described* (F1 23.3 to 32.5 vs. 19.5 to 31.4), and neither reaches the published 33.3 on average (§7). We also find that the published USR numbers correspond to an evaluation window that keeps the *end* of each review, whereas the released code keeps the beginning (§6). Because 85% of the test reviews exceed BERT's 512-token limit, this choice alone changes F1 by 1.7 to 3.6 points.

The analysis points to the capacity of the generator rather than to the noise. We therefore keep the generator's encoder at its pretrained weights, so that only a linear token scorer and a linear classifier (2,307 parameters) are trained, and we rank tokens by their attention averaged over a five-token window, so that the selection budget is spent on phrases instead of isolated tokens (§5). Neither change adds a training objective. Against our reproductions, the frozen generator is at least as plausible as NI as described in all three seeds (by 0.3 to 11.9 F1 points), more plausible than NI as released in two of three seeds and by 3.8 points on average, more plausible than the fine-tuned generator without noise, and far more stable. Phrase selection adds two to three points to every generator. Under the published protocol, the result improves on the best published model in IOU-F1 (50.9 vs. 38.4) and accuracy (91.8 vs. 87.9) and slightly in token F1 (34.2 vs. 33.3), and it trains in about two thirds of the time (§7). In the window of the released code, it reaches F1 31.0, against 16.4–24.7 for our fine-tuned reproductions trained there.

Our contributions are:

- an analysis showing that noise injection does not steer the generator of BERT-A2R away from the words it replaces, with four diagnostics: the exposure of the selection to noise, its share of stopwords, the accuracy of attention restricted to stopwords, and the accuracy of the predictor on random selections;
- a reproducibility audit of the USR benchmark: a model-independent test that identifies the evaluation window from published precision and recall, and fixes for four defects in the released code, one of which changes the F1 of NI by -4 to $+10$ points;
- a frozen generator, which without any noise is more plausible on average than noise injection, as described or as released, and far more stable across seeds; and phrase selection, which adds two to three F1 points to any generator; together they improve on the published state of the art on USR Movie Review under the published protocol;
- negative results for adaptive and evidence-path noise.

We release our code and all diagnostics.[1]

2 Related Work

Selective rationalization. Lei et al. (2016) train a generator and an encoder jointly with REINFORCE and penalties for sparsity and continuity. Later work makes selection differentiable (Bastings et al., 2019), controls conciseness with an information bottleneck (Paranjape et al., 2020), or treats generator and predictor as players of a cooperative game (Chen et al., 2018; Chang et al., 2019, 2020). A2R (Yu et al., 2021) adds an attention-based predictor that gives the generator a dense gradient, and BERT-A2R (Storek et al., 2023) replaces A2R's GloVe-based generator with BERT. Degeneration has been traced to the interplay of the two players: Yu et al. (2019) observe generators that communicate the label through punctuation, Jacovi and Goldberg (2021) argue that any generator able to represent the label can encode it in its selection, and Zheng et al. (2022) question how rational rationale models are. Proposed remedies include a shared encoder (Liu et al., 2022), a lower learning rate for the generator (Liu et al., 2023b), multiple generators (Liu et al., 2023a), diagnostic objectives (Atanasova et al., 2022), and noise injection (Storek et al., 2023). Freezing the generator's encoder is the limit of the asymmetric learning rates of Liu et al. (2023b); on USR, this limit is more plausible on average than a fine-tuned generator with noise injection (§7). Supervised alternatives learn from annotated rationales (Zaidan et al., 2007; DeYoung et al., 2020) or train the components separately (Jain et al., 2020); we stay unsupervised.

[1] [TODO: anonymized repository link]

Contiguous rationales. Humans annotate phrases. Earlier models encourage contiguity with a fused-lasso penalty (Lei et al., 2016; Bastings et al., 2019), structured inference (Guerreiro and Martins, 2021), or by extracting contiguous spans (Jain et al., 2020). Our phrase selection is a parameter-free smoothing of the attention before top-k selection and does not change the training objective.

Frozen representations and attention. Frozen pretrained features are competitive with fine-tuning on many tasks (Peters et al., 2019), and a linear head on frozen states is a probe whose capacity is limited by design (Hewitt and Liang, 2019). Whether attention weights explain a model's predictions is debated (Jain and Wallace, 2019; Wiegreffe and Pinter, 2019); in selective rationalization the attention only decides what the predictor sees, so the rationale is faithful regardless.

Evaluation. We follow ERASER (DeYoung et al., 2020) for plausibility and faithfulness metrics and Carton et al. (2020) for normalized faithfulness; Jacovi and Goldberg (2020) discuss the definitions. Sun et al. (2019) show that the part of a long review that BERT keeps affects its sentiment accuracy; we show that it also changes rationale metrics, so published numbers can only be compared within the same window. Our audit adds to work on reproducibility in NLP (Belz et al., 2021).

3 Background: BERT-A2R with Noise Injection

Let $x = (x_1, \ldots, x_T)$ be a tokenized review with label y. BERT-A2R (Storek et al., 2023) has three parts (Figure 1).

Generator. A BERT encoder maps x to states $h_1, \ldots, h_T$. A linear scorer computes $s_i = w^\top h_i$, and $a = \mathrm{softmax}(s)$ is the generator's attention.

Attention-based predictor. It shares the generator's encoder and classifies from the attention-pooled states, $\hat{y}^a = \mathrm{softmax}(W \sum_i a_i h_i)$.

Selection and rationale-based predictor. The rationale mask m keeps the $k = \mathrm{round}(0.2\,T)$ tokens with the highest attention. A second BERT reads the selected tokens in their original order and predicts $\hat{y}^r$.

Training. Each predictor minimizes its cross-entropy plus a closeness term:

$$\mathcal{L}_a = \mathrm{CE}(\hat{y}^a, y) + \lambda\,\mathrm{JSD}\left(\hat{y}^a, \mathrm{sg}[\hat{y}^r]\right), \quad (1)$$

$$\mathcal{L}_r = \mathrm{CE}(\hat{y}^r, y) + \lambda\,\mathrm{JSD}\left(\mathrm{sg}[\hat{y}^a], \hat{y}^r\right), \quad (2)$$

where sg stops the gradient and $\lambda = 0.1$. The generator receives gradient only from Eq. 1.

Noise injection. During training, each selected word x_i is replaced, before the rationale-based predictor reads it, with probability $\min(1, p\,r_i)$ by a word drawn from the vocabulary with a probability that decreases with its ATF∗IDF, a scheme adapted from unsupervised data augmentation (Xie et al., 2020). The replacement weight

$$r_i = |d|\,\frac{t_{\max} - t_i}{\sum_{j \in d}(t_{\max} - t_j)} \quad (3)$$

is high for words with low TF∗IDF t_i in document d. By construction, r has mean 1 over a document, so a randomly selected word is replaced with probability p on average. Storek et al. (2023) use $p = 0.2$ on USR.

4 What Noise Injection Does Not Do

4.1 No token-level credit assignment

Eq. 1 is the only loss that reaches the generator. The rationale-based predictor enters it through $\mathrm{sg}[\hat{y}^r]$, a two-class distribution for the whole document weighted by $\lambda = 0.1$, and top-k selection is not differentiable. When noise replaces a selected word, the generator can therefore learn at most that $\hat{y}^r$ changed, not which word was replaced or that selecting it was a mistake. The mechanism NI relies on, making implausible selections costly for the generator, has no path in this architecture.

This yields a direct test. If NI penalized the selection of the words it targets, the generator would learn to avoid them. We measure the *exposure* of a rationale, the mean replacement weight r_i (Eq. 3) of its words: it is 1 for a random selection and below 1 for a selection that avoids noise targets. In every run with NI, the exposure stays at about 1.0 throughout training (1.02 at the end of training in both seeds of the frozen generator of §5 with NI, and 1.02–1.08 in the three seeds of the fine-tuned generator trained in TAIL): noise hits the generator's choices at least as often as it would hit random ones.

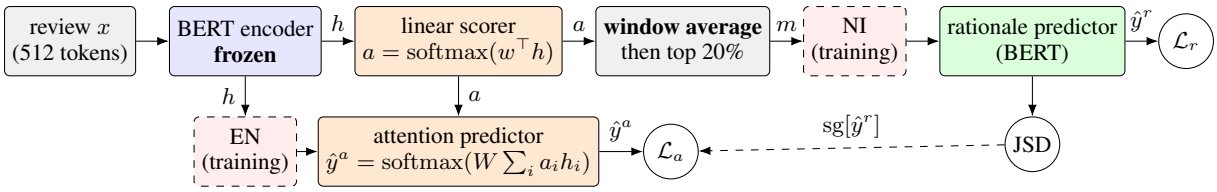


Figure 1: BERT-A2R and our changes (bold). Orange: the generator, trained only by $\mathcal{L}_a$ (Eq. 1); green: the rationale predictor, trained by $\mathcal{L}_r$; blue: frozen in our model. The rationale predictor reaches $\mathcal{L}_a$ only through a stop-gradient JSD term (dashed), and top-k selection passes no gradient, so noise on the rationale (NI) cannot tell the generator which selected word was replaced. Noise on the generator's evidence (EN, §5.3) enters $\mathcal{L}_a$ directly. We freeze the encoder and average the attention over a window before selection.

Model	Acc	P	R	F1	IOU	Stop
Fine-tuned generator, trained in HEAD						
A2R	88.1	11.2	30.6	16.4	17.8	.64
+ NI	85.1	13.4	36.7	19.6	20.5	.56
+ $\mathrm{NI}_{\mathrm{released}}$	81.7	16.8	46.1	24.7	27.3	.61
+ $\mathrm{NI}_{\mathrm{PI}}$	81.3	11.6	31.8	17.0	17.5	.71
Fine-tuned generator, trained in TAIL						
A2R	86.4	12.1	33.1	17.7	17.6	.65
+ NI	85.0	14.7	40.1	21.5	24.0	.63
+ $\mathrm{NI}_{\mathrm{released}}$	84.7	16.8	45.8	24.5	27.6	.63
Frozen generator, trained in TAIL						
FG	87.1	19.0	52.0	27.8	27.0	.41
FG + PS	88.4	21.1	57.8	**31.0**	**46.8**	.51*
Random selection	–	7.3	20.0	10.7	–	.51
Human rationales	–	–	–	–	–	.46

Table 1: Evaluation in HEAD, the window of the released code (fine-tuned: seed 1 in HEAD, mean over three seeds in TAIL; frozen: mean over three seeds). Stop: share of stopwords and punctuation among the selected words. $\mathrm{NI}_{\mathrm{released}}$: NI with the alignment defect of the released code (App. A); $\mathrm{NI}_{\mathrm{PI}}$: noise level set by a controller on predictor disagreement. FG: frozen generator; PS: phrase selection (§5). Random selection: expected scores of selecting 20% of the tokens at random (stopword share measured in TAIL). Human rationales: all annotated words of the test reviews. *Phrases include the function words between content words, as human rationales do.

4.2 Fine-tuned generators select function words

Table 1 shows our reproduction of BERT-A2R with a fine-tuned generator under the protocol of the released code. Trained in HEAD, it reaches token F1 between 16.4 and 24.7 across noise settings, and every run spends more of its rationale on stopwords and punctuation (56–71%) than a random selection of the same size (51%), the human annotators (46%), or a frozen generator (41%). Fixed NI raises F1 from 16.4 to 19.6 in this single seed without lowering the exposure; NI as released, whose rationale predictor receives the token *before* each selected token (App. A), reaches 24.7. A closed-loop controller that raised p whenever the two predictors disagreed on the clean rationale had nothing to react to: they agreed even when 71% of the rationale consisted of function words. Trained in HEAD, none of these runs comes close to the published fine-tuned results; trained in TAIL, A2R+NI and NI as released each do in one of three seeds (§7), yet even these runs spend 63% and 56% of their selections on stopwords and punctuation.

4.3 A contextual encoder can place the label on any token

Why does the generator not prefer the evidence? Its only objective, Eq. 1, asks the attention-pooled states to predict the label. Each state of a deep contextual encoder mixes information from the whole review, so the label is available at almost any position. We verify this with generator heads trained on cached BERT-base states (App. E): restricting the attention to stopwords and punctuation leaves accuracy unchanged for pretrained states (81.8% vs. 81.8%) and even raises it for the states of a BERT fine-tuned on IMDb (90.3% vs. 89.1%). A fine-tuned scorer can also encode the label in *which* tokens it selects: when it reads fine-tuned states but pools non-contextual (layer-0) embeddings as evidence, it still reaches 89.3% accuracy while 57% of its selection are stopwords (F1 18.2). This is the communication described by Yu et al. (2019) and Jacovi and Goldberg (2021). Rationale noise does not reach the place where the code is formed; at most, it makes the code unreadable for the predictor (§8). Even noise moved into the generator's own evidence (§5.3) does not help a fine-tuned encoder (App. E). What a fine-tuned generator selects depends on its capacity to form such codes, which motivates freezing it.

5 Method

5.1 Frozen generator (FG)

We keep the generator's encoder at its pretrained BERT-base weights and train only the scorer w and the classifier W of the attention-based predictor ($769 + 1{,}538 = 2{,}307$ parameters). The rationale-based predictor is fine-tuned as before, and the losses (Eqs. 1–2) are unchanged. With frozen states, the scorer is a linear probe (Hewitt and Liang, 2019): it ranks tokens along one direction of a fixed representation space and has far less freedom to form a code than a fine-tuned encoder. Empirically, it ranks content words highest: on average, only 41% (HEAD, Table 1) and 42% (TAIL) of its single-token selections are stopwords, fewer than in a random selection (51%) or in human rationales (46%). Storek et al. (2023) report this configuration as their fixed-generator baseline; with matched evaluation, it is more plausible on average than our reproductions of their best model, the fine-tuned generator with NI, with or without the defect of the released code (§7).

5.2 Phrase selection (PS)

Top-k selection of single tokens produces scattered tokens: the frozen generator's selections form spans of 1.9 words on average, while human rationales consist of spans of 7.7 words. We rank tokens by their attention averaged over a centered window of w tokens,

$$\tilde{a}_i = \frac{1}{|N_i|} \sum_{j \in N_i} a_j, \qquad (4)$$

where $N_i = \{j : |j-i| \le (w-1)/2,\ 1 \le j \le T\}$, and select the k tokens with the highest $\tilde{a}_i$. The budget k is unchanged. A token with high attention lifts its neighbors, so the budget is spent on phrases around the evidence. Phrase selection has no parameters, is applied in the same way during training and evaluation, and can be applied to any trained generator. We use $w = 5$, shorter than the average human span; §7.2 reports $w \in \{1, 5, 9\}$.

5.3 Evidence noise (EN), an ablation

To test whether noise helps once it can reach the generator, we also move it into the generator's own evidence. During training, each state h_i in the pooled sum of the attention-based predictor is replaced, with probability $\min(1, p\, r_i)$, by the (detached) state of a word drawn uniformly from the batch; the attention a is still computed from the clean states. Attending to a word that the noise targets now yields unreliable evidence, which shows up in $\mathcal{L}_a$ and thus in the generator's gradient. Its effect is measured by the *attention exposure* $E = \sum_i a_i r_i$, which is 1 for uniform attention. We also tried to hold E at a target with a proportional-integral controller on p (App. E).

6 Experimental Setup

Data. USR Movie Review (Storek et al., 2023) trains on 40,000 IMDb reviews (Maas et al., 2011), validates on 10,000 more, and tests on the 2,000 reviews of ERASER Movie Reviews (DeYoung et al., 2020; Zaidan and Eisner, 2008; Pang and Lee, 2004), which carry token-level human rationales. The test reviews are long (774 words on average), and 85% of them exceed 512 wordpieces. Human rationales cover 7.6% of the words, in 7.6 spans of 7.7 words per review; 46% of their words are stopwords or punctuation.

Evaluation window. For a review longer than 512 wordpieces, BERT sees either its beginning (HEAD, the released code) or its end (TAIL). The choice matters for plausibility: 54% of the human-annotated words survive truncation in HEAD and 68% in TAIL, as reviewers tend to state their verdict at the end. Which window do the published numbers use? Every model selects the same fraction of the tokens, so micro precision and recall share their numerator and their ratio $R/P = S/G$ is the number of selected tokens divided by the number of annotated tokens inside the window: a property of the window, not of the model. All our evaluations give $R/P = 2.735$ in HEAD and 2.164–2.165 in TAIL; all four USR rows of Storek et al. (2023) give 2.16 (App. D). The published numbers therefore correspond to TAIL, and we compare with them in TAIL only, reporting HEAD as well. As in the released code, plausibility is computed over the tokens inside the window.

Metrics. Following Storek et al. (2023), we report token-level micro precision, recall and F1 against the human rationales, IOU-F1 with partial-match threshold 0.1 (DeYoung et al., 2020; Paranjape et al., 2020), and the accuracy of the rationale-based predictor. For faithfulness, we report comprehensiveness and sufficiency (DeYoung et al., 2020) of the rationale-based predictor. The released code applies a sigmoid to these proba-

bilities, which compresses all differences into $[0.5, 0.73]$; we report this scale for comparability and the scores on actual probabilities. Because a predictor trained on rationales is out of distribution on full reviews, we also score faithfulness with an independent BERT classifier trained on full reviews, using its probability for the class it predicts from the full review: comprehensiveness and sufficiency at the trained selection rate, both normalized by the empty input (Carton et al., 2020), and their area over the perturbation curve (AOPC) over selection rates of 1, 5, 10, 20 and 50% (DeYoung et al., 2020). As diagnostics, we report a random selection of the same size, the share of stopwords and punctuation among the selected words, and the mean span length.

Models. *Published:* the four BERT-A2R variants of Storek et al. (2023) (fixed or fine-tuned generator, with or without NI). *Reproduced:* BERT-A2R with a fine-tuned generator without noise (A2R), with NI at $p = 0.2$ as described, i.e. after fixing the alignment defect (+NI), with NI as released (+$\text{NI}_{\text{released}}$), and with NI whose level is set by a controller on predictor disagreement (+NI_{PI}), all trained in HEAD with seed 1; A2R, +NI and +$\text{NI}_{\text{released}}$ are also trained in TAIL with seeds 1 to 3. *Ours:* the frozen generator (FG), with phrase selection (FG+PS), and with noise on the rationale (+NI, $p = 0.2$) or on the evidence (+EN, $p = 0.3$).

Implementation. We use `bert-base-uncased` with Transformers (Wolf et al., 2020) and the settings of Storek et al. (2023): Adam (Kingma and Ba, 2015) with learning rate $2 \cdot 10^{-5}$ for both components, batch size 16, at most 5 epochs, $k = 20\%$ and $\lambda = 0.1$. We keep the checkpoint of the last epoch in which both validation losses improved (patience 2); validation is noise-free. Unlike Storek et al. (2023), who train the fixed generator with batch size 64 for 10 epochs, we use the same settings for all models. Each run uses one NVIDIA T4 GPU (16 GB). We report FG+PS and the fine-tuned models trained in TAIL as means over three seeds and the noise ablations over seeds 1 and 2. Differences are computed from unrounded means. Training is deterministic given the seed: re-running FG+PS with seed 2 in another session reproduced every training loss and every per-review result, so two models trained with the same seed differ only by their configuration. We compare such pairs with a paired bootstrap over the 2,000 test reviews (10,000 resamples) and report 95% confidence intervals. The independent classifier is a BERT-base fine-tuned on the training reviews in TAIL with the same optimizer, batch size and early stopping (seed 3). Unlike the rationalization models, its training was not exactly repeatable across sessions: the two instances we trained reach 94.0% and 93.7% test accuracy.

Model	Acc	P	R	F1	IOU
Published (Storek et al., 2023), 5 seeds					
Fixed gen.	$85.0_{0.8}$	21.9	47.4	$30.0_{0.5}$	$29.9_{0.6}$
Fixed gen. + NI	$85.8_{1.1}$	22.3	48.2	$30.5_{0.6}$	$30.7_{0.8}$
Tuned gen.	$82.4_{8.6}$	20.2	43.7	$27.6_{3.0}$	$29.1_{5.3}$
Tuned gen. + NI	$87.9_{1.8}$	24.4	52.7	$33.3_{0.8}$	$38.4_{1.9}$
Reproduced, tuned generator, trained in TAIL					
A2R	$89.1_{1.8}$	14.2	30.7	$19.4_{3.4}$	$19.0_{2.8}$
+ PS‡	$80.3_{12.2}$	16.4	35.5	$22.4_{2.5}$	$33.9_{3.5}$
+ NI	$88.6_{3.4}$	17.9	38.8	$24.5_{6.2}$	$27.1_{7.2}$
+ NI + PS‡	$86.4_{5.0}$	19.8	42.8	$27.0_{6.7}$	$41.4_{9.8}$
+ $\text{NI}_{\text{released}}$	$87.1_{4.4}$	20.2	43.7	$27.6_{4.6}$	$30.9_{4.8}$
+ $\text{NI}_{\text{released}}$ + PS‡	$88.4_{2.7}$	22.1	47.9	$30.3_{5.1}$	$46.6_{6.5}$
Ours, frozen generator					
FG†	$90.9_{1.1}$	23.0	49.8	$31.5_{0.2}$	$30.5_{0.5}$
FG + PS	$91.8_{0.4}$	25.0	54.2	$34.2_{0.1}$	$50.9_{0.3}$
Random selection	$74.1_{0.6}$	9.2	20.0	$12.6_{0.1}$	$7.6_{0.1}$

Table 2: USR Movie Review, TAIL window (the protocol of the published numbers, §6). Subscripts: standard deviation. Ours and reproduced: mean over three seeds. $\text{NI}_{\text{released}}$: NI with the alignment defect of the released code (App. A). Random selection: the FG+PS predictor given a random selection of the same size. †The FG+PS model evaluated with single-token selection; a frozen generator trained with single tokens (seed 3) has the same F1 (31.7) and a 1.2 points more accurate predictor than FG† of the same seed. ‡Evaluated with phrase selection ($w = 5$), which the model was not trained with.

7 Results

7.1 Main results

Under the published protocol (Table 2), the frozen generator alone is at least as plausible as the published fixed-generator row (F1 31.5 vs. 30.0, IOU-F1 30.5 vs. 29.9). Phrase selection adds 2.8 F1 and 20.5 IOU-F1 points. FG+PS improves on the best published model, the fine-tuned generator with NI, by 0.9 F1, 12.5 IOU-F1 and 3.9 accuracy points. The F1 margin is about one published standard deviation and not significant (Welch's t-test on the published means and standard deviations: $p = 0.06$); the margins in IOU-F1 ($p < 0.001$) and accuracy ($p = 0.007$) are. The rationale matters to the predictor: replacing it with a random

selection of the same size lowers accuracy from 91.8 to 74.1.

Trained and evaluated in TAIL, our reproductions of the fine-tuned generator vary strongly between seeds: with NI, F1 is 19.5, 22.5 and 31.4 in seeds 1 to 3, close to the published 33.3 only in seed 3; without noise, it is 17.2, 17.7 and 23.3 (published: 27.6). Paired by seed (Table 5 in App. B), NI raises F1 in every seed, by 5.1 points on average. With single tokens, the frozen generator is 12.1 F1 points more plausible than A2R and 7.0 points more plausible than A2R+NI on average; it is more plausible than both in every seed, except that it only matches A2R+NI in seed 3 ($+0.3$, not significant). Its F1 varies little between seeds (31.3 to 31.7). Phrase selection adds about three F1 points to every generator. FG+PS improves on A2R+NI by 9.8 F1 points on average and in every seed, with a predictor that is 3.2 points more accurate on average; in seed 3, the gain comes from phrase selection alone (A2R+NI+PS: $+0.5$ over FG+PS).

NI as released varies almost as much between seeds: F1 23.3, 32.5 and 27.1 in seeds 1 to 3 (Table 2), close to the published 33.3 only in seed 2. Paired by seed (Table 6), the defect changes F1 by $+3.8$, $+10.0$ and -4.3 points. NI as released is more plausible than A2R in every seed, but less plausible than the frozen generator in seeds 1 and 3 (by 8.1 and 4.6 points; seed 2: -1.3) and than FG+PS in every seed (by 6.6 points on average).

In the HEAD window (Table 1), FG+PS, trained in TAIL, outperforms every fine-tuned reproduction trained in HEAD by 6.3–14.6 F1 points (31.0 vs. 16.4–24.7). Trained in TAIL, A2R+NI reaches 28.3 there in seed 3 but only 16.6 and 19.5 in seeds 1 and 2, and NI as released 20.2 to 29.3.

7.2 Ablations

Window and selection. The two factors add up (Table 3): TAIL adds 3.3–3.6 F1 points and phrase selection 2.8–3.1. Spans grow from 1.9 words ($w = 1$) to 5.9 ($w = 5$) and 9.1 ($w = 9$). IOU-F1 grows with the window, from 30.5 to 50.9 and 60.4, while F1 peaks at $w = 5$ (34.2) and drops at $w = 9$ (33.3). With a 0.1 threshold, IOU-F1 credits every generated span that overlaps a human span by a tenth, so it rewards contiguity as such; we therefore treat token F1 as the primary plausibility metric. Accuracy is also higher with phrases (91.8 vs. 90.9), although part of this gap may come from evaluating, with single tokens, a predictor that was trained on phrases; accuracy with single tokens also varies more between seeds (standard deviation 1.1 vs. 0.4).

Window	Selection	Acc	P	R	F1	IOU
HEAD	$w = 1$	87.1	19.0	52.0	27.8	27.0
HEAD	$w = 5$	88.4	21.1	57.8	31.0	46.8
TAIL	$w = 1$	90.9	23.0	49.8	31.5	30.5
TAIL	$w = 5$	**91.8**	**25.0**	**54.2**	**34.2**	50.9
TAIL	$w = 9$	91.1	24.3	52.6	33.3	**60.4**
TAIL	random	74.1	9.2	20.0	12.6	7.6

Table 3: The FG+PS model evaluated under each window and selection (mean over three seeds; standard deviation at most 0.2 for F1, 0.5 for IOU-F1 and 1.1 for accuracy; bold: best in TAIL). The window and the selection add up; IOU-F1 keeps growing with w while F1 peaks at $w = 5$.

Window versus model. The gain of TAIL over HEAD is a protocol choice, not a method gain. It differs between models (1.7 F1 points for A2R, 3.0 to 3.6 for the others) and can be as large as the published gain of the fine-tuned generator with NI over the fixed generator (3.3 F1), so the window must be fixed before comparing numbers.

7.3 Does noise help a frozen generator?

Neither form of noise improves plausibility (Table 7 in App. C). Paired by seed, NI changes the F1 and IOU-F1 of FG+PS by less than 0.1 points, as §4.1 predicts. Noise on the evidence does reach the generator: its attention exposure falls to 0.83, and its single-token selections contain fewer stopwords. Yet its F1 changes by $+0.3$ and -0.4 points, and its accuracy falls by about one point.

7.4 Faithfulness

With the rationale-based predictor (Table 4, top), the comprehensiveness of FG+PS lies in the published range (.034 vs. .02–.04 on the sigmoid scale), and its sufficiency is closer to zero ($-.007$ vs. $-.01$ to $-.04$). Negative sufficiency, i.e. more confidence on the rationale than on the full review, is expected from a predictor trained only on rationales and makes these scores hard to compare across models. The independent classifier (bottom) is not trained on rationales. Keeping only the rationale of FG+PS lowers the classifier's probability for the class it predicts by .045 on average, and the rationale alone retains 88% of the gain in this probability over an empty input (on the 97% of the reviews where there is a gain). Removing the rationale lowers the probability by

	Sigmoid scale		Probabilities	
Model	Com ↑	Suf ↓	Com ↑	Suf ↓
Published (Storek et al., 2023)				
Fixed gen.	.02	−.02	–	–
Fixed gen. + NI	.03	−.01	–	–
Tuned gen.	.03	−.03	–	–
Tuned gen. + NI	.04	−.04	–	–
Reproduced, trained in TAIL				
A2R	.022	−.062	.091	−.270
A2R + NI	.033	−.056	.140	−.247
Ours				
FG + PS	.034	−.007	.157	−.037
+ NI	.033	−.012	.151	−.056
+ EN	.032	−.008	.143	−.040
Independent classifier				
	Probabilities		Normalized	
	Com ↑	Suf ↓	Com ↑	Suf ↑
A2R + NI, reproduced	.335	.029	.56	.90
FG + PS	.175	.045	.35	.88

Table 4: Faithfulness in TAIL. Top: comprehensiveness and sufficiency of each model's own rationale-based predictor (sigmoid scale: as computed by the released code; FG+PS and reproduced: mean over three seeds; +NI, +EN: seeds 1 and 2). Bottom (seed 3): the same scores from an independent classifier trained on full reviews, raw and normalized by its score on an empty input (Carton et al., 2020). Each model is scored by a classifier trained in its own session with the same settings and seed (94.0% and 93.7% test accuracy).

.175, or 35% of that gain. Over selection rates from 1% to 50%, the AOPC is .135 for comprehensiveness and .089 for sufficiency. Our reproduction of the fine-tuned generator with NI in its best seed (seed 3), scored by a second classifier trained with the same settings and seed (93.7% accuracy), is twice as comprehensive (.335; AOPC .228) and slightly more sufficient (.029; normalized .90). Only its AOPC sufficiency is worse (.117), because its smallest selections are less sufficient. A fine-tuned generator selects what a fine-tuned classifier relies on, including the stopwords and punctuation (63% of its single-token selections) that make its rationales less plausible.

7.5 Cost

Freezing removes all but 2,307 of the generator's 109.5M trainable parameters and halves the number of parameters that receive gradients in BERT-A2R. On a T4 GPU, an epoch takes 0.7–0.9 hours with the frozen generator and 1.3–1.8 hours with a fine-tuned one (run times: Table 11).

8 Discussion

What noise injection does. NI does not tell the generator which of its choices were implausible (§4.1), yet it made our fine-tuned generators more plausible in all four paired runs, by 3.3 F1 points in HEAD and by 2.4 to 8.1 in TAIL. Given a random selection of the same size, the predictor of A2R assigns more than 98% of the reviews to one class in all three seeds: it reads a code in the generator's choice of tokens, not their content. Noise on the selected words can break such a code, close to the explanation of Storek et al. (2023) that the noise disrupts the generator's model of the predictor, but not always: the predictor of A2R+NI reads no code only in seed 3 (71.1% accurate on random selections), and that of NI as released, which is trained on the word before each selected token, only in seeds 2 and 3 (67.9%). Across the nine fine-tuned runs trained in TAIL, plausibility follows the code: F1 is 17.2 to 23.3 in the six runs whose predictor assigns more than 90% of random selections to one class, and 27.1 to 32.5 in the other three. Freezing prevents the code in every seed (FG+PS: 73.4–74.6%), and NI then has no measurable effect (Table 7).

Recommendations for evaluation. Rationale metrics on long documents depend on the truncation window as much as on the model. We recommend reporting the window, the plausibility of a random selection of the same size, the stopword share of the selection, and the predictor's accuracy on random selections. The ratio R/P is a cheap check that two tables use the same window and selection rate.

9 Conclusion

Noise injection was proposed to keep powerful generators from producing implausible rationales. In BERT-A2R, it does not do so by steering the generator: the noise has no token-level path to it, and the selected words are as exposed to noise as random ones. Its benefit varies strongly between seeds, with or without a defect of the released code that shows the predictor the word before each selected token, and the rationales are plausible only when the predictor no longer reads a code in the selection. Freezing the generator's encoder prevents the code in every seed without noise; its rationales are more plausible on average than with noise injection, as described or as released, and vary far

less. Phrase selection adds about three F1 points to any generator; together, the two improve on the published state of the art. That state of the art was measured on the end of each review, a choice that matters as much as the method. We hope these results shift attention from perturbing the generator's training to limiting what a generator can encode in its choice of tokens.

Limitations

We evaluate on a single benchmark of English movie reviews with one encoder, BERT-base; whether pretrained states localize evidence as well in other domains, languages and encoders is untested, and tasks with sentence-level rationales such as MultiRC and FEVER remain open. Our main model and the fine-tuned models trained in TAIL are averaged over three seeds, the noise ablations over two, and the runs trained in HEAD and the independent classifiers are single runs (the classifier's training was not exactly repeatable across sessions); differences below one point, such as the effect of evidence noise or the F1 margin over the best published model, are within seed variation. Our fine-tuned reproductions vary far more between seeds than the published runs and fall short of them on average, with or without the alignment defect of the released code (App. A): F1 19.5 to 31.4 (mean 24.5) as described and 23.3 to 32.5 (mean 27.6) as released, against 33.3 with a published standard deviation of 0.8. We do not know what else differed in the published runs. USR has no annotated development set, so the diagnostic experiments that informed the window size and noise level used a subset of the annotated test reviews; to limit the effect of this choice, we report every window we tried. Because of BERT's 512-token limit, the windows lose 46% (HEAD) and 32% (TAIL) of the human rationales; long-document encoders would remove this confound. The selection budget is fixed at 20% of the tokens, while the amount of evidence varies between reviews. Finally, we measure plausibility as agreement with human annotations, which are themselves inconsistent between reviews (Storek et al., 2023); we did not run a human evaluation.

Ethics Statement

Selective rationales are faithful by construction, but plausible rationales can still make users overtrust a model; a rationale shows what the model used, not that the decision is right. Our models are trained on public movie-review datasets that contain no personal information beyond what reviewers chose to publish. Reducing the trainable parameters of the generator lowers the compute needed to train rationalization models.

References

Pepa Atanasova, Jakob Grue Simonsen, Christina Lioma, and Isabelle Augenstein. 2022. Diagnostics-guided explanation generation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(10):10445–10453.

Jasmijn Bastings, Wilker Aziz, and Ivan Titov. 2019. Interpretable neural predictions with differentiable binary variables. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2963–2977, Florence, Italy. Association for Computational Linguistics.

Anya Belz, Shubham Agarwal, Anastasia Shimorina, and Ehud Reiter. 2021. A systematic review of reproducibility research in natural language processing. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 381–393, Online. Association for Computational Linguistics.

Samuel Carton, Anirudh Rathore, and Chenhao Tan. 2020. Evaluating and characterizing human rationales. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9294–9307, Online. Association for Computational Linguistics.

Shiyu Chang, Yang Zhang, Mo Yu, and Tommi Jaakkola. 2019. A game theoretic approach to class-wise selective rationalization. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

Shiyu Chang, Yang Zhang, Mo Yu, and Tommi Jaakkola. 2020. Invariant rationalization. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 1448–1458. PMLR.

Jianbo Chen, Le Song, Martin Wainwright, and Michael Jordan. 2018. Learning to explain: An information-theoretic perspective on model interpretation. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 883–892. PMLR.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association*

for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. 2020. ERASER: A benchmark to evaluate rationalized NLP models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4443–4458, Online. Association for Computational Linguistics.

Nuno M. Guerreiro and André F. T. Martins. 2021. SPECTRA: Sparse structured text rationalization. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6534–6550, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

John Hewitt and Percy Liang. 2019. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Hong Kong, China. Association for Computational Linguistics.

Alon Jacovi and Yoav Goldberg. 2020. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4198–4205, Online. Association for Computational Linguistics.

Alon Jacovi and Yoav Goldberg. 2021. Aligning faithful interpretations with their social attribution. *Transactions of the Association for Computational Linguistics*, 9:294–310.

Sarthak Jain and Byron C. Wallace. 2019. Attention is not Explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3543–3556, Minneapolis, Minnesota. Association for Computational Linguistics.

Sarthak Jain, Sarah Wiegreffe, Yuval Pinter, and Byron C. Wallace. 2020. Learning to faithfully rationalize by construction. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4459–4473, Online. Association for Computational Linguistics.

Diederik P. Kingma and Jimmy Ba. 2015. Adam: A method for stochastic optimization. In *3rd International Conference on Learning Representations*.

Tao Lei, Regina Barzilay, and Tommi Jaakkola. 2016. Rationalizing neural predictions. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 107–117, Austin, Texas. Association for Computational Linguistics.

Wei Liu, Haozhao Wang, Jun Wang, Ruixuan Li, Xinyang Li, YuanKai Zhang, and Yang Qiu. 2023a. MGR: Multi-generator based rationalization. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12771–12787, Toronto, Canada. Association for Computational Linguistics.

Wei Liu, Haozhao Wang, Jun Wang, Ruixuan Li, Chao Yue, and Yuankai Zhang. 2022. FR: Folded rationalization with a unified encoder. In *Advances in Neural Information Processing Systems*, volume 35. Curran Associates, Inc.

Wei Liu, Jun Wang, Haozhao Wang, Ruixuan Li, Yang Qiu, Yuankai Zhang, Jie Han, and Yixiong Zou. 2023b. Decoupled rationalization with asymmetric learning rates: A flexible Lipschitz restraint. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 1535–1547. ACM.

Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. 2011. Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pages 142–150, Portland, Oregon, USA. Association for Computational Linguistics.

John Morris, Eli Lifland, Jin Yong Yoo, Jake Grigsby, Di Jin, and Yanjun Qi. 2020. TextAttack: A framework for adversarial attacks, data augmentation, and adversarial training in NLP. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 119–126, Online. Association for Computational Linguistics.

Bo Pang and Lillian Lee. 2004. A sentimental education: Sentiment analysis using subjectivity summarization based on minimum cuts. In *Proceedings of the 42nd Annual Meeting of the Association for Computational Linguistics (ACL-04)*, pages 271–278, Barcelona, Spain.

Bhargavi Paranjape, Mandar Joshi, John Thickstun, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2020. An information bottleneck approach for controlling conciseness in rationale extraction. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1938–1952, Online. Association for Computational Linguistics.

Matthew E. Peters, Sebastian Ruder, and Noah A. Smith. 2019. To tune or not to tune? adapting pretrained representations to diverse tasks. In *Proceedings of the 4th Workshop on Representation Learning for NLP (RepL4NLP-2019)*, pages 7–14, Florence, Italy. Association for Computational Linguistics.

Adam Storek, Melanie Subbiah, and Kathleen McKeown. 2023. Unsupervised selective rationalization

with noise injection. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12647–12659, Toronto, Canada. Association for Computational Linguistics.

Chi Sun, Xipeng Qiu, Yige Xu, and Xuanjing Huang. 2019. How to fine-tune BERT for text classification? In *Chinese Computational Linguistics*, volume 11856 of *Lecture Notes in Computer Science*, pages 194–206. Springer.

Sarah Wiegreffe and Yuval Pinter. 2019. Attention is not not explanation. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 11–20, Hong Kong, China. Association for Computational Linguistics.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Qizhe Xie, Zihang Dai, Eduard Hovy, Thang Luong, and Quoc V. Le. 2020. Unsupervised data augmentation for consistency training. In *Advances in Neural Information Processing Systems*, volume 33. Curran Associates, Inc.

Mo Yu, Shiyu Chang, Yang Zhang, and Tommi Jaakkola. 2019. Rethinking cooperative rationalization: Introspective extraction and complement control. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4094–4103, Hong Kong, China. Association for Computational Linguistics.

Mo Yu, Yang Zhang, Shiyu Chang, and Tommi S. Jaakkola. 2021. Understanding interlocking dynamics of cooperative rationalization. In *Advances in Neural Information Processing Systems*, volume 34. Curran Associates, Inc.

Omar Zaidan and Jason Eisner. 2008. Modeling annotators: A generative approach to learning from annotator rationales. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, pages 31–40, Honolulu, Hawaii. Association for Computational Linguistics.

Omar Zaidan, Jason Eisner, and Christine Piatko. 2007. Using "annotator rationales" to improve machine learning for text categorization. In *Human Language Technologies 2007: The Conference of the North American Chapter of the Association for Computational Linguistics; Proceedings of the Main Conference*, pages 260–267, Rochester, New York. Association for Computational Linguistics.

Yiming Zheng, Serena Booth, Julie Shah, and Yilun Zhou. 2022. The irrationality of neural rationale models. In *Proceedings of the 2nd Workshop on Trustworthy Natural Language Processing (TrustNLP 2022)*, pages 64–73, Seattle, U.S.A. Association for Computational Linguistics.

A Defects in the Released Code

We found four defects in the released implementation of Storek et al. (2023) and fixed them in our code; each fix can be switched off to reproduce the original behavior.

1. The script that computes word statistics crashed before writing its output.
2. Training with noise required a file of replacement probabilities for the validation set that no script created.
3. **Alignment.** When noise was injected, the word indices of the selected tokens were computed with the `[CLS]` token still in place, so the rationale-based predictor received the word of the token *preceding* each selected token. This affects training only; validation and test use the selected tokens. With the same seed, the defect changed the F1 of NI from 19.6 to 24.7 in HEAD (seed 1, Table 1) and, in TAIL, from 19.5, 22.5 and 31.4 to 23.3, 32.5 and 27.1 in seeds 1 to 3 (Table 6). We call NI with the fix *NI as described* (+NI) and NI with the defect *NI as released* (+$\text{NI}_{\text{released}}$).
4. **Faithfulness.** Comprehensiveness and sufficiency applied a sigmoid to values that were already probabilities, which maps every probability into $[0.5, 0.73]$. We report both scales (Table 4).

The pinned tokenizer of the released code truncates long reviews from the right, i.e. it keeps the beginning (HEAD).

B Paired Comparisons

Table 5 compares models of Table 2 that were trained with the same seed. NI makes the fine-tuned generator more plausible in all three seeds, but by different amounts: in seeds 1 and 2, A2R+NI remains far less plausible than the frozen generator; in seed 3, it is as plausible, and FG+PS

	Seed 1		Seed 2		Seed 3		Mean
Difference	Δ	95% CI	Δ	95% CI	Δ	95% CI	Δ
Token F1							
A2R+NI − A2R	2.37	1.93, 2.82	4.85	4.51, 5.20	8.09	7.34, 8.85	5.10
FG − A2R	14.27	13.67, 14.88	13.58	13.00, 14.18	8.37	7.68, 9.06	12.08
FG − A2R+NI	11.91	11.16, 12.65	8.73	8.13, 9.36	0.28	−0.11, 0.67	6.97
A2R+PS − A2R	4.65	4.31, 4.98	2.57	2.28, 2.87	1.89	1.68, 2.09	3.03
A2R+NI+PS − A2R+NI	2.27	2.05, 2.50	2.11	1.90, 2.31	3.26	3.05, 3.47	2.55
FG+PS − FG	2.95	2.70, 3.19	2.91	2.66, 3.16	2.48	2.23, 2.73	2.78
FG+PS − A2R+NI	14.86	14.07, 15.65	11.65	10.98, 12.32	2.76	2.32, 3.21	9.76
FG+PS − A2R+NI+PS	12.58	11.68, 13.49	9.54	8.80, 10.30	−0.50	−0.98, −0.01	7.21
Accuracy							
A2R+NI − A2R	−1.95	−3.30, −0.60	−1.50	−3.10, 0.10	1.85	0.55, 3.15	−0.53
FG − A2R	1.45	0.00, 2.95	4.55	2.90, 6.20	−0.85	−2.20, 0.55	1.72
FG − A2R+NI	3.40	1.85, 4.95	6.05	4.45, 7.70	−2.70	−4.05, −1.40	2.25
A2R+PS − A2R	−23.50	−26.00, −20.95	−1.80	−3.20, −0.45	−1.25	−2.50, 0.00	−8.85
A2R+NI+PS − A2R+NI	−4.30	−6.20, −2.35	−2.25	−3.90, −0.55	−0.15	−0.65, 0.35	−2.23
FG+PS − FG	0.90	−0.20, 1.95	0.15	−1.00, 1.30	1.80	0.75, 2.90	0.95
FG+PS − A2R+NI	4.30	2.85, 5.75	6.20	4.70, 7.75	−0.90	−2.15, 0.40	3.20
FG+PS − A2R+NI+PS	8.60	6.90, 10.25	8.45	6.70, 10.20	−0.75	−2.00, 0.50	5.43

Table 5: Paired differences between models trained with the same seed, evaluated in TAIL (percentage points; 95% confidence intervals from a paired bootstrap over the 2,000 test reviews with 10,000 resamples; Mean: over the three seeds). FG: the FG+PS model evaluated with single-token selection. +PS on a fine-tuned model: phrase selection at evaluation only. The accuracy differences of these rows and of FG include a predictor evaluated with a selection it was not trained with.

owes its gain over A2R+NI to phrase selection. Applied at evaluation, phrase selection costs the fine-tuned predictors accuracy, most for A2R in seed 1 (23.5 points). In HEAD (seed 1, Table 1), NI raises F1 by 3.27 points (2.10 to 4.43) and lowers accuracy by 2.95 points (1.30 to 4.65).

Table 6 compares NI as released with the other models trained with the same seed. The defect changes F1 by -4.3 to $+10.0$ points and accuracy by -9.0 to $+6.5$ points. NI as released is more plausible than A2R in every seed and less plausible than the frozen generator in seeds 1 and 3. FG+PS is more plausible than NI as released in every seed, and than NI as released with phrase selection in seeds 1 and 3. In HEAD (seed 1, both trained in HEAD), the defect raises F1 by 5.03 points (4.64 to 5.42) and lowers accuracy by 3.40 points (1.55 to 5.20).

C Noise on the Frozen Generator

Table 7 adds noise to FG+PS. Because training is deterministic given the seed (§6), each noise run differs from FG+PS with the same seed only by the noise. In both seeds, FG+PS+NI reaches the F1 of FG+PS with the same seed to within 0.04 points and its IOU-F1 to within 0.07, while the two seeds of FG+PS differ by 0.2 and 0.6 points. NI changes only the accuracy of the rationale-based predictor, by -0.85 and -0.05 points. Noise on the evidence does reach the generator: its attention exposure falls to 0.83 in both seeds, against 0.89–0.92 in a run whose controller kept the noise level at 0.04–0.06, and in both seeds its single-token selections contain fewer stopwords than without noise (38% vs. 42% and 33% vs. 38%). Yet its F1 changes by $+0.3$ and -0.4 points in the two seeds, and its accuracy falls by 0.80 and 0.95 points. On cached states (App. E), the useful range of the evidence-noise level is narrow: F1 improves by about one point up to $p = 0.5$, degrades at $p = 0.7$ and collapses at $p = 1$. A controller that held the attention exposure at 0.9 matched the best fixed level there, but in the full pipeline the generator's exposure already sits near that target without noise, and the controller kept the noise level at 0.06 or below.

D Identifying the Evaluation Window

With the same selection rate for every model, micro precision $P = \mathrm{TP}/S$ and recall $R = \mathrm{TP}/G$, where S is the number of selected tokens and G the number of annotated tokens inside the window.

	Seed 1		Seed 2		Seed 3		Mean
Difference	Δ	95% CI	Δ	95% CI	Δ	95% CI	Δ
Token F1							
A2R+NI$_{rel}$ − A2R+NI	3.79	3.46, 4.13	9.99	9.39, 10.59	−4.32	−4.74, −3.90	3.15
A2R+NI$_{rel}$ − A2R	6.16	5.73, 6.58	14.84	14.27, 15.40	3.77	2.93, 4.61	8.26
FG − A2R+NI$_{rel}$	8.12	7.47, 8.76	−1.26	−1.60, −0.90	4.60	4.15, 5.05	3.82
A2R+NI$_{rel}$+PS − A2R+NI$_{rel}$	1.97	1.77, 2.18	2.86	2.67, 3.05	3.11	2.89, 3.32	2.65
FG+PS − A2R+NI$_{rel}$	11.07	10.39, 11.75	1.66	1.24, 2.07	7.08	6.59, 7.59	6.60
FG+PS − A2R+NI$_{rel}$+PS	9.09	8.33, 9.87	−1.20	−1.64, −0.76	3.97	3.44, 4.53	3.95
Accuracy							
A2R+NI$_{rel}$ − A2R+NI	−1.95	−3.40, −0.45	6.45	4.90, 8.05	−8.95	−10.75, −7.15	−1.48
A2R+NI$_{rel}$ − A2R	−3.90	−5.35, −2.45	4.95	3.30, 6.60	−7.10	−9.00, −5.20	−2.02
FG − A2R+NI$_{rel}$	5.35	3.65, 7.05	−0.40	−1.55, 0.80	6.25	4.25, 8.30	3.73
A2R+NI$_{rel}$+PS − A2R+NI$_{rel}$	1.25	−0.05, 2.55	−0.60	−1.30, 0.10	3.10	2.20, 4.05	1.25
FG+PS − A2R+NI$_{rel}$	6.25	4.70, 7.85	−0.25	−1.45, 0.95	8.05	6.30, 9.80	4.68
FG+PS − A2R+NI$_{rel}$+PS	5.00	3.50, 6.55	0.35	−0.85, 1.55	4.95	3.35, 6.55	3.43

Table 6: Paired differences involving NI as released (A2R+NI$_{rel}$, with the alignment defect), evaluated in TAIL as in Table 5. A2R+NI is NI as described.

Model	Acc	P	R	F1	IOU	Stop
FG + PS	92.0	25.1	54.3	34.3	50.8	.40
+ NI ($p = 0.2$)	91.6	25.1	54.3	34.3	50.8	.42‡
+ EN ($p = 0.3$)	91.1	25.0	54.1	34.2	50.1	.35

Table 7: Noise on top of FG+PS, TAIL. Every row is the mean over seeds 1 and 2, so each noise run is compared with FG+PS trained with the same seed (FG+PS over three seeds: Table 2). Stop: stopword share of single-token selections, i.e. where the generator's attention peaks. ‡Seed 1 only.

Hence

$$\frac{R}{P} = \frac{S}{G} \approx \frac{0.2}{\rho}, \tag{5}$$

where ρ is the share of annotated tokens in the window. The ratio does not depend on the model. On the test set, $\rho = 7.3\%$ in HEAD and 9.2% in TAIL, so $R/P \approx 2.73$ and 2.16. Table 8 compares the measured ratios with the published ones.

Source	P	R	R/P
Published			
Fixed gen.	21.9	47.4	2.16
Fixed gen. + NI	22.3	48.2	2.16
Tuned gen.	20.2	43.7	2.16
Tuned gen. + NI	24.4	52.7	2.16
Ours			
HEAD (38)	10.5–21.7	28.8–59.4	2.735
TAIL (69)	9.2–25.9	19.9–56.0	2.164–2.165

Table 8: The ratio of recall to precision identifies the window. In parentheses: number of our evaluations, which span fine-tuned and frozen generators, single tokens, phrases and random selections.

E Diagnostic Experiments on Cached States

To test generator variants quickly, we cached the hidden states of BERT-base for 2,000 training reviews and 1,000 randomly drawn annotated test reviews, in both windows, from layers 0, 4, 8 and 12 of two encoders: the pretrained `bert-base-uncased` and a BERT fine-tuned on IMDb (`textattack/bert-base-uncased-imdb`, Morris et al., 2020). On these states we trained the generator's two linear heads (scorer and attention-based classifier) with Adam (learning rate $2 \cdot 10^{-3}$, batch size 32, 20 epochs); no rationale-based predictor is involved, and accuracy is that of the attention-based predictor. Results are means over 2 seeds (Table 9) or 3 seeds (Table 10). The setup also predicts the full pipeline: trained with its learning rate ($2 \cdot 10^{-5}$) and number of updates (12,500), with one seed, the heads predicted F1 31.1 and 34.4 in TAIL for single tokens and phrases, against 31.5 and 34.2 measured (mean over three seeds), and IOU-F1 30.3 and 51.0, against 30.5 and 50.9.

The label is available everywhere. Table 9 shows that the attention-based predictor classifies as well from stopwords and punctuation as from all tokens. With free attention, heads on pretrained layer-12 states select single tokens of which 38% are stopwords (F1 26.3 in HEAD), and phrase selection raises F1 to 30.1. Heads on fine-tuned states select 51% stopwords, as many as a random selection (F1 25.6), and phrase selection barely

	Attention allowed on		
Layer-12 states	all tokens	stopwords	content words
Pretrained	81.8	81.8	82.2
Fine-tuned on IMDb	89.1	90.3	90.3

Table 9: Accuracy of the attention-based predictor when its attention is restricted to stopwords and punctuation or to content words (HEAD, 2 seeds). The label is available at any position. [TODO: raw output not preserved; re-run before submission or drop.]

helps (26.1): their attention peaks are not localized on the evidence. Lower layers of the pretrained encoder give lower F1 (14.5, 18.3, 23.2 and 26.3 for layers 0, 4, 8 and 12). A scorer on fine-tuned layer-12 states that pools layer-0 embeddings as evidence reaches 89.3% accuracy with 57% stopwords and F1 18.2, i.e. it encodes the label in its choice of tokens.

How much the rationale can carry. The IMDb-fine-tuned classifier reaches 95.4% accuracy on the human rationales alone and 89.3% on the full reviews (HEAD), on the same 1,000 test reviews: a plausible rationale is enough to classify, so better rationales can also raise accuracy. [TODO: raw output not preserved; re-run before submission or drop.]

		HEAD, $w = 1$		TAIL, $w = 5$		
p	E	F1	Acc	F1	IOU	Acc
0	.92	$26.1_{0.4}$	81.8	$33.5_{0.4}$	48.7	86.7
0.1	.94	$26.4_{0.2}$	83.5	$33.4_{0.3}$	49.3	88.7
0.2	.95	$26.9_{0.3}$	84.2	$33.8_{0.2}$	49.6	89.2
0.3	.95	$27.3_{0.2}$	85.0	$34.3_{0.1}$	51.3	89.4
0.5	.94	$27.3_{0.5}$	84.8	$34.6_{0.3}$	50.9	89.2
0.7	.87	$25.1_{0.3}$	83.0	$33.2_{0.5}$	48.4	87.9
1.0	.59	$14.1_{0.9}$	72.6	$17.1_{0.8}$	24.6	77.2
PI, $E^* = 0.9$	.92	$27.3_{0.4}$	85.2	$34.6_{0.1}$	50.6	88.9
PI, $E^* = 0.8$	.79	$21.9_{0.8}$	80.9	$30.3_{0.7}$	43.5	85.8
PI, $E^* = 0.7$	.68	$18.3_{0.3}$	76.2	$24.9_{0.3}$	35.7	83.7

Table 10: Evidence noise on frozen pretrained states (3 seeds; subscripts: standard deviation). E: attention exposure at the end of training. PI: a proportional-integral controller that sets p to hold E at the target E^*.

Evidence noise. Table 10 varies the level of evidence noise on the frozen pretrained states. A mild level ($p = 0.3$–0.5) adds about one F1 point and three accuracy points; stronger noise collapses the attention onto the words the noise spares. A controller that holds the exposure at 0.9 finds the best fixed level without a sweep, but lower targets are clearly worse. On fine-tuned states, evidence noise does not help ($p = 0$: F1 24.9; $p = 0.3$: 24.9; $p = 1$: 14.9).

F Training Details

Encoder	`bert-base-uncased`
Optimizer	Adam, learning rate $2 \cdot 10^{-5}$
Batch size	16
Epochs, patience	5, 2
Selection rate k	20%
Closeness λ	0.1
Maximum length	512 wordpieces
Window w	5
Noise level	NI: $p = 0.2$; EN: $p = 0.3$
Trainable parameters	generator: 2,307 (FG) or 109.5M (fine-tuned); rationale predictor: 109.5M
Independent classifier	`bert-base-uncased` on full reviews in TAIL, same optimizer, batch size, epochs and patience; 94.0% test accuracy
Hardware	one NVIDIA T4 (16 GB) per run
Time per run	FG: 3.5–4.6 h (0.7–0.9 h per epoch); fine-tuned: 4.2–8.2 h (1.3–1.8 h per epoch)

Table 11: Hyperparameters and compute.

Table 11 lists our settings. Training and evaluation ran on Kaggle notebooks with two T4 GPUs, one run per GPU. [TODO: total GPU hours.] We use BERT (Apache 2.0), the USR Movie Review benchmark (MIT), ERASER Movie Reviews (Apache 2.0) and the IMDb reviews of Maas et al. (2011), as listed by Storek et al. (2023), all for research purposes consistent with their release. [TODO: license of the TextAttack model.]